

Advanced pancreatic cancer: is there a role for combination therapy?

PMID: 14599095 | Indexed in PubMed

Once thought to be a relatively untreatable disease, pancreatic cancer has recently become a focus of intense clinical research. The systemic administration of gemcitabine (Gemzar) is currently considered the standard first-line treatment for patients with advanced disease. While treatment with gemcitabine has been shown to result in both clinical benefit and prolongation of survival, objective tumor responses are relatively uncommon and median survival times remain short. Several recent efforts have therefore focused on evaluating chemotherapy regimens in which gemcitabine is combined with other cytotoxic drugs. While randomized trials have now confirmed that such combinations are associated with higher response rates, they have not yet clearly demonstrated that combination therapy results in a survival advantage. Increasingly, attention has turned to a number of novel chemotherapeutic and biologic agents that appear promising and are likely to play an important future role in the treatment of patients with this disease.

State of the art biological therapies in pancreatic cancer.

PMID: 26798437 | Indexed in PubMed

Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal malignancies with a five-year survival rate of approximately 5%. Several target agents have been tested in PDAC, but almost all have failed to demonstrate efficacy in late phase clinical trials, despite the better understanding of PDAC molecular biology generated by large cancer sequencing initiatives in the past decade. Erlotinib (a small-molecule tyrosine-kinase inhibitor of epidermal growth factor receptor) plus gemcitabine is the only schedule with a biological agent approved for advanced pancreatic cancer, but it has resulted in a very modest survival benefit in unselected patients. In our work, we report a summary of the main clinical trials (closed and ongoing) that refer to biological therapy evaluation in pancreatic cancer treatment.

A TRAILR2/CDH3 bispecific antibody demonstrates selective apoptosis and tumor regression in CDH3-positive pancreatic cancer.

PMID: 39654063 | Indexed in PubMed

Exploitation of extrinsic apoptosis signaling via TRAILR2 activation represents a promising therapeutic concept in cancer treatment. The limited clinical success of previous TRAILR2 agonistic agents, to date, has been ascribed to either poor efficacy or hepatotoxicity. TR2/CDH3 BAB is a human bispecific antibody that relies on binding both CDH3 and TRAILR2 on cell surfaces to achieve TRAILR2 hyperclustering and efficient apoptosis induction by TRAILR2 signaling selectively in CDH3-expressing tumor cells. We demonstrate target-dependent TR2/CDH3 BAB anti-tumor activity in CRISPR/Cas9-engineered TRAILR2 or CDH3 knock-out cells. By utilizing the cell line screening platform PRISM, we found selective TR2/CDH3 BAB efficacy in various cancer types, such as pancreatic, gastric, colorectal, and triple negative breast cancer. The efficacy of TR2/CDH3 BAB correlated with caspase activation in cancer cell lines and in xenograft tumor tissues. In pancreatic ductal adenocarcinoma (PDAC), where patient benefit from current cytotoxic therapy options is unsatisfactory, a close to uniform cell surface expression of CDH3 and TRAILR2 was observed, which will qualify the majority of PDAC patients for TR2/CDH3 BAB-based

treatment. TR2/CDH3 BAB demonstrated anti-tumor activity in a panel of PDAC patient-derived xenograft models, including tumor regressions. By combining TR2/CDH3 BAB with chemotherapeutic agents, deeper and more sustained anti-tumor responses were observed when compared to monotherapy. Together with the potential to deliver a favorable safety profile, these data support clinical testing of TR2/CDH3 BAB in patients with PDAC.

Erlotinib: the first biologic in the management of pancreatic cancer.

PMID: 18518788 | Indexed in PubMed

Pancreatic cancer remains a major therapeutic challenge largely characterized by chemotherapy-refractory disease and poor responses to currently available treatments. Pivotal studies of combination chemotherapeutic agents with gemcitabine showed that responses may exceed single-agent gemcitabine, but with added toxicities. The EGFR is known to be overexpressed in pancreatic cancer and data suggest that the presence of EGFR is associated with a poor prognosis. To analyze the current data on erlotinib and compare against other agents targeting against the EGFR pathway in pancreatic cancer. PubMed, Ovid, Cochran and the American Society of Clinical Oncology abstract database were searched using the terms 'erlotinib', 'EGFR', 'cetuximab', 'gemcitabine', 'fluorouracil', 'capecitabine' and 'pancreatic cancer' to identify relevant studies. Only studies using erlotinib in pancreatic cancer patients were reviewed and analyzed. The combination of gemcitabine and erlotinib is the first combination therapy to demonstrate survival benefits in pancreatic cancer in a Phase III study albeit a modest one. Increased understanding of the EGFR pathway may permit the use of other targeted agents to either augment therapeutic efficacy or circumvent resistance. It is warranted to develop strategies to truly target our therapy with the EGFR agents by identifying those patients who are most likely to derive benefit and achieve meaningful responses.

Cetuximab: still an option in the treatment of pancreatic cancer?

PMID: 23560505 | Indexed in PubMed

In this review, we analyzed the current literature about cetuximab to clarify its role in the treatment of pancreatic cancer. Using single-agent gemcitabine has been the standard treatment for more than 15 years for advanced pancreatic cancer. The attempts at improving the results by combining it with several other drugs, such as fluorouracil, cisplatin, irinotecan, oxaliplatin, or pemetrexed produced no clear survival benefit. Recently, however, new combination chemotherapy regimens (e.g., FOLFIRINOX, nab-paclitaxel plus gemcitabine) achieved a significant survival benefit compared to gemcitabine alone. Epidermal growth factor receptor (EGFR) transmembrane glycoprotein has been demonstrated to be overexpressed in pancreatic cancer, and it correlates with more advanced disease, poor survival, and the presence of metastases. Therefore, inhibition of the EGFR signaling pathway could be an attractive therapeutic target in this tumor. Although several combinations of EGFR inhibitors with chemotherapy demonstrate inhibition of tumor-induced angiogenesis, tumor cell apoptosis, and regression in xenograft models, these benefits remain to be confirmed. The encouraging results from preclinical and early clinical studies with cetuximab in pancreatic cancer were not confirmed in a Phase III trial. Cetuximab failed to demonstrate improved patient outcome when paired with various chemotherapeutic regimens and/or other biological agents.

Biologic therapies for advanced pancreatic cancer.

PMID: 18699769 | Indexed in PubMed

Patients with metastatic pancreatic cancer have poor prognosis and short survival due to lack of effective therapy and aggressiveness of the disease. Pancreatic cancer has widespread chromosomal instability, including a high rate of translocations and deletions. Upregulated EGF signaling and mutation of K-RAS are found in most pancreatic cancers. Therefore, inhibitors that target EGF receptor, K-RAS, RAF, MEK, mTOR, VEGF and PDGF, for example, have been evaluated in patients with pancreatic cancer. Although significant activities of these inhibitors have not been observed in the majority of pancreatic cancer patients, an enormous amount of experience and knowledge has been obtained from recent clinical trials. With a better inhibitor or combination of inhibitors, and improvement in the selection of patients for available inhibitors, better therapy for pancreatic cancer is on the horizon.

Adjuvant therapy of pancreatic cancer using monoclonal antibodies and immune response modifiers.

PMID: 9127173 | Indexed in PubMed

Pancreatic cancer is a devastating disease with poor survival. At present, no effective adjuvant or palliative therapies are available. Unresponsiveness to chemotherapy, radiotherapy, and antihormonal treatment is one of the reasons that pancreatic cancer patients have an overall median survival time of 4-6 mo. This article summarizes clinical trials on immunotherapy of pancreatic cancer using the murine monoclonal antibodies (MAbs) 17-1A and BW 494. In addition, the use of MAb treatment in combination with immune response modifiers is discussed. In four clinical trials, MAb 17-1A was given by iv infusion to 100 patients with pancreatic cancer. In 30 of the these patents, antibody treatment was accompanied by gamma-interferon, also given intravenously. Complete response, partial response, and stable disease were reported in 1, 5, and 23 patients, respectively. Passive immunotherapy using the MAb BW 494 was carried out in 148 pancreatic cancer patients in two phase I and two phase II trials. In 1 out of 75 patients a partial response and in 25 out of 74 patients stable disease were reported. However, in a controlled randomized trial enrolling 61 patients following Whipple resection, comparable survival times in patients with and without MAb BW 494 treatment led to the termination of further clinical trials with this antibody. New clinical studies using humanized MAbs in combination with immune response modifiers should be initiated to further evaluate immunotherapy as a treatment option in pancreatic cancer.

Monoclonal antibodies and other targeted therapies for pancreatic cancer.

PMID: 23187854 | Indexed in PubMed

Pancreatic cancer continues to be a challenging disease to treat because of its aggressive nature, advanced stage at the time of diagnosis, and limited treatment options that are available. Traditional cytotoxic chemotherapy provides modest benefit to patients with pancreatic adenocarcinoma. Recently, a FOLFIRINOX regimen revealed improved response in overall and progression-free survival over single-agent gemcitabine in metastatic pancreatic cancer, but there is still much needed advancement in

the systemic treatment of pancreatic cancer. There is a growing interest in the development of novel agents, while our understanding of molecular pathogenesis of pancreatic adenocarcinoma continues to expand. With identification of various molecular pathways in pancreatic cancer tumorigenesis, potential targets for drug development have been pursued with the use of monoclonal antibodies and small-molecule inhibitors. Although preclinical studies with multiple targeted therapies demonstrated encouraging results in pancreatic cancer, only erlotinib, an epidermal growth factor receptor inhibitor, showed a marginal survival benefit in a phase III clinical trial, when combined with gemcitabine. As further signaling pathways and their importance in pancreatic cancer tumorigenesis are better understood, further clinical trials will need to be designed to study these targeted agents as single agents, in combination with other novel agents or in combination with cytotoxic chemotherapy. In this review, we present the current knowledge on targeted therapy in pancreatic adenocarcinoma and its application in clinical practice.

Human pancreatic cancer-associated antigens detected by murine monoclonal antibodies.

PMID: 4063969 | Indexed in PubMed

Two monoclonal antibodies, YPan1 and YPan2, were produced from spleen cells of mice immunized against a human pancreatic cancer cell line, Capan-2, which also reacted with eight other human pancreatic carcinoma cell lines and with sections of cancerous human pancreas tissue. Reactivity was also found with colonic and stomach carcinoma tissues. Whereas YPan1 reacted strongly with normal pancreatic tissue, it bound weakly, if at all, to a variety of other normal tissues. YPan1 reacted with both the membranes and cytoplasm of pancreatic adenocarcinoma cells and with constituents of normal pancreatic ductal cells and duct luminal contents. YPan2, on the other hand, reacts to a greater degree with intracytoplasmic constituents in pancreatic adenocarcinoma cells. Unlike YPan1, YPan2 reacted weakly with only one of 16 cases of normal pancreas. Pretreatment of the tissue with neuraminidase abolished YPan1's reactivity, which indicated that sialic acid may be involved in the antigenic activity of the molecule(s) recognized by YPan1. Neuraminidase pretreatment had no effect on YPan2 reactivity. Neither YPan1 nor YPan2 competed with 19-9 monoclonal antibody in binding to soluble CA 19-9 antigen. These results suggest that these monoclonal antibodies are of potential use in the diagnosis of pancreatic carcinoma.

Characterization of monoclonal antibody PAM4 reactive with a pancreatic cancer mucin.

PMID: 7512537 | Indexed in PubMed

A monoclonal antibody (MAb), PAM4, having reactivity with pancreatic carcinoma has been developed. PAM4 is an IgG1 immunoglobulin produced by immunization of mice with mucin purified from the xenografted RIP1 human pancreatic carcinoma. An immunohistochemical study of normal adult tissues showed the PAM4 reactive epitope to be restricted to the gastrointestinal tract and absent from normal pancreas. In neoplastic tissue, PAM4 was reactive with 85% of the pancreatic carcinomas, approximately half of the colon cancers and none of the breast, ovarian, prostate, renal and liver cancers. PAM4 was, in general, non-reactive with pancreatitis specimens whereas CA19.9 and DUPAN2 were strongly reactive with each one. Treatment of the mucin antigen by heating, reduction of disulfide bonds, or protease

digestion abolished immunoreactivity with PAM4. Treatment of the mucin by neuraminidase or periodate oxidation reduced immunoreactivity but did not completely abolish it. Our data are consistent with the proposal that the PAM4 epitope is a conformationally dependent peptide epitope and that certain carbohydrate structures are necessary in order to maintain the correct peptide conformation. The high specificity and intense reactivity of PAM4 with pancreatic carcinoma tissue suggests that the antibody may prove useful for in vitro diagnostic assays as well as in vivo targeting of diagnostic and therapeutic agents.
